# 📘 MODULE 21: UNIVERSAL FORWARDER

### 21.1 Concept

The **Universal Forwarder (UF)** is a lightweight, separate Splunk package (not the full Splunk Enterprise) installed on the machine that produces the logs. It reads data and sends it to an indexer. It has **no Splunk Web** and does no search.

### 21.2 Why it matters

Nearly all production data arrives via UFs. "A UF isn't sending data" is the most common onboarding ticket and a favorite interview scenario.

### 21.3 Key facts

- Default install: `/opt/splunkforwarder` (Linux), `C:\Program Files\SplunkUniversalForwarder` (Windows).
- Windows service: `SplunkForwarder`.
- Sends data to the indexer's **receiving port (9997)**. Its own management port is 8089.
- Does minimal processing. Exception: structured data (`INDEXED_EXTRACTIONS`) can be parsed on a UF.
- Sends its own logs to `_internal` too, so you can monitor UFs from the indexer.
- **Lab warning:** if the UF is on the same machine as your Splunk Enterprise, port 8089 conflicts. The UF will offer another port, so accept it. Better: use a second Ubuntu VM.

### 21.4 Install and configure (Linux), step by step

**Step 1: install** (run in the folder containing the downloaded `.deb`)

```bash
sudo dpkg -i splunkforwarder-<version>-linux-amd64.deb
```

`dpkg -i` installs the package into `/opt/splunkforwarder`.

**Step 2: first start**

```bash
sudo /opt/splunkforwarder/bin/splunk start --accept-license
```

It asks you to create an admin username and password for the UF (local only).

**Step 3: point it at the indexer**

```bash
sudo /opt/splunkforwarder/bin/splunk add forward-server 192.168.1.10:9997 -auth admin:<pas
```

- `add forward-server` writes the target into `outputs.conf`.
- `192.168.1.10:9997` is the indexer's IP and receiving port. **Use your indexer's IP.**

- Expected: `Added forwarding to: 192.168.1.10:9997.`

#### Step 4: add a monitor

```bash
sudo /opt/splunkforwarder/bin/splunk add monitor /var/log/auth.log -index linux_auth -sou
```

- Writes a `[monitor://...]` stanza to `inputs.conf`.
- The `linux_auth` index **must exist on the indexer**, or the data is dropped.

**Step 5: permissions.** The UF user must read the file (see Module 2.7, `adm` group).

#### Step 6: boot start and restart

```bash
sudo /opt/splunkforwarder/bin/splunk enable boot-start
sudo /opt/splunkforwarder/bin/splunk restart
```

### 21.5 The same thing as files

`/opt/splunkforwarder/etc/system/local/outputs.conf`

```ini
[tcpout]
defaultGroup = my_indexers

[tcpout:my_indexers]
server = 192.168.1.10:9997
```

`/opt/splunkforwarder/etc/system/local/inputs.conf`

```ini
[monitor:///var/log/auth.log]
index = linux_auth
sourcetype = linux_secure
disabled = 0
```

Save, then `sudo /opt/splunkforwarder/bin/splunk restart`.
Optional in `[tcpout:...]`: `useACK = true` (indexer acknowledgment to avoid data loss) and several servers (`server = a:9997, b:9997`) for automatic load balancing.

### 21.6 The receiving end (indexer)

Settings → Forwarding and receiving → Configure receiving → New Receiving Port → `9997`. Or `splunk enable listen 9997 -auth admin:<pw>`.

### 21.7 Windows UF (Splunk perspective only)

Install the `.msi`, set the receiving indexer in the wizard, and choose which logs to collect. Config is the same `inputs.conf` / `outputs.conf` under `C:\Program Files\SplunkUniversalForwarder\etc\system\local\`. The CLI is `"C:\Program Files\SplunkUniversalForwarder\bin\splunk.exe" status`. Service control: `net stop SplunkForwarder` and `net start SplunkForwarder`. Example input:

```ini
[WinEventLog://Security]
index = windows
disabled = 0
renderXml = false
```

Windows event channels are only readable if the UF service account has rights (the default Local System account usually does).

### 21.8 Deployment Server (concept)

Managing 500 UFs by hand is impossible. A **Deployment Server (DS)** holds **deployment apps** and **server classes** (which clients get which apps). Each UF has a `deploymentclient.conf`:

```ini
[target-broker:deploymentServer]
targetUri = ds.example.com:8089
```

The UF periodically phones home and downloads its apps (an `inputs.conf` bundle, for example). On the DS: Settings → Forwarder Management. Interview point: **the DS is not in the data path, it only distributes config.**

### 21.9 Verify and test connectivity

| Where | Command | Meaning |
|---|---|---|
| UF | `splunk status` | Is the UF running |
| UF | `splunk list forward-server` | **Active** = connected, **configured but inactive** = not connected |
| UF | `splunk list monitor` | Files being monitored |
| UF | `splunk btool outputs list --debug` | Effective outputs |
| UF | `splunk list inputstatus` | Per-file read progress |
| UF | `nc -zv 192.168.1.10 9997` | Network reachability to the port |
| UF | `tail -50 /opt/splunkforwarder/var/log/splunk/splunkd.log` | UF errors |
| Indexer | `index=_internal source=*metrics.log* group=tcpin_connections \| stats latest(_time) by hostname, sourceIp` | Did the indexer see this UF? |
| Indexer | `index=_internal host=<uf-host> \| head` | UF's own logs arriving |
| Indexer | `index=linux_auth host=<uf-host> \| head` | The actual data |

### 21.10 Troubleshooting

| Symptom | Likely cause and fix |
|---|---|
| `list forward-server` shows inactive | Wrong IP/port in `outputs.conf`, receiver not enabled, firewall. Test with `nc`. |
| Active but no data | Input not reading (permissions, wrong path), index doesn't exist on indexer, or no new lines (monitor reads only new content after the first read) |
| `Permission denied` in UF log | UF user can't read the file |
| Data in `main` not expected index | Input missing `index=`, or another stanza wins (`btool --debug`) |
| Duplicate data | UF re-installed/fishbucket reset, or two inputs on the same file |
| UF visible in `_internal` but not data | Data path problem, not a connectivity problem |

### 21.11 Interview Q&A

- **What is a UF?** A lightweight agent that collects and forwards data with minimal processing, with no UI and no indexing.
- **UF vs HF?** UF is light, no web UI, can't route or filter by content; HF is a full Splunk instance that can parse, filter, mask, and route.
- **Two key config files on a UF?** `inputs.conf` (what) and `outputs.conf` (where).
- **UF running, no data. What do you check?** `list forward-server`, port reachability, UF `splunkd.log`, indexer `tcpin_connections`, index existence, permissions, time range.
- **How do you manage many UFs?** Deployment Server with server classes.
- **Does the UF parse data?** Mostly no. Parsing happens at the indexer or HF, with exceptions such as structured data.

### 21.12 Revision

UF = light agent · `inputs.conf` + `outputs.conf` · receiver 9997 · `list forward-server` · check `_internal` on the indexer · DS manages config, not data.